The Analysis and Documentation Framework

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THE ANALYSIS AND DOCUMENTATION FRAMEWORK

PfE 5106 Enterprise Architecture

Start

LEARNING OUTCOMES

At the end of this lesson, students should be able to:

1. Explain the role of analysis and documentation in Enterprise Architecture
2. Distinguish analysis activities from documentation activities in EA
3. Identify key principles that guide effective EA analysis and documentation
4. Describe how EA frameworks conceptually support analysis and documentation
5. Recognize why analysis and documentation are required before As-Is and To-Be architecture views

Lesson Content

A. What Is the Analysis & Documentation Framework in EA
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What Is the Analysis & Documentation Framework in EA

Enterprise Architecture (EA) is not just conceptual thinking; it is a disciplined practice that managers and architects use to understand, plan, and govern enterprise change. In professional EA practice, architecture work always begins with ***analysis and documentation*** — not modelling or diagrams.

Analysis is the process of examining enterprise elements, identifying relationships and gaps, and structuring knowledge logically.

Documentation is the process of capturing, organizing, storing, and communicating the results of analysis so that it supports planning, decision-making, governance, and future evolution of the enterprise.

Why EA Requires Structured Analysis

Enterprise Architecture replaces fragmented thinking with ***enterprise-wide analysis***.

According to the industry-recognized TOGAF Standard (The Open Group), architecture analysis supports alignment between:

- Business strategy
- Business processes
- Information needs
- Technology resources

Without structured analysis:

- Architects make decisions based on intuition
- Different units execute incompatible plans
- Architects overlook dependencies that cause project failure

What “Analysis” Means in EA Practice

In EA practice, analysis means more than reading documents or creating lists. It involves:

1. **Scoping** – defining what parts of the enterprise will be analyzed
2. **Data gathering** – collecting information about business processes, systems, technology, and goals
3. **Comparative analysis** – identifying gaps, duplications, overlaps, redundancies
4. **Dependency analysis** – understanding how one part influences another
5. **Impact analysis** – examining how changes in one area affect others
6. **Stakeholder analysis** – identifying who cares about what and why

Professional EA methods structure these steps so that data is not only collected but ***interpreted*** — turning information into knowledge.

Why Documentation Matters in EA

Documentation is not just notes; ***it is the formal record of architecture knowledge*** that can be:

- **Reviewed**
- **Stored**
- **Shared**
- **Reused**
- **Queried**
- **Tracked over time**

The most credible EA methodologies (e.g., TOGAF, ArchiMate) treat documentation as ***living knowledge***, not static files.

Good EA Documentation:

- Uses a common vocabulary
- Follows standard formats
- Relates architecture elements consistently
- Is organized for quick retrieval
- Includes metadata (who, when, why, and which version)

This is an important difference between ***documents and documentation***:

- A *document* may be a report or diagram
- *Documentation* is a ***systematic collection of architecture knowledge***

Key Principles of EA Analysis and Documentation

Professional EA standards define principles that must guide analysis and documentation:

- **Completeness** Architecture must not ignore any major element of the enterprise.
- **Consistency** All documentation must use consistent definitions and relationships.
- **Traceability** Every documented item should trace back to requirements or decisions.
- **Reusability** Analysis artifacts should be reusable in planning and change.
- **Maintainability** Documentation must be updatable as the enterprise evolves.

These principles help enterprise architects produce documentation that is ***actionable, auditable, and sustainable.***

Frameworks Guide Analysis & Documentation

Enterprise Architecture frameworks DO NOT do the analysis for you. Instead, they provide ***structures for how analysis and documentation should be done***.

For Example:

TOGAF

TOGAF defines phases that require analysis and documentation:

- Preliminary (scope & principles)
- Architecture Vision (high-level requirements)
- Business, Information, and Technology Architecture stages

At each phase, requirements are analyzed and documented before moving forward.

Zachman Framework

Zachman does not prescribe steps but defines perspectives and questions:

- What (data)
- How (function)
- Where (network)
- Who (people)
- When (time)
- Why (motivation)

These categories guide what must be analyzed and documented.

Archimate

ArchiMate provides a visual language for documenting relationships across layers (business, application, technology) in a way that supports traceability and clarity.

Types of EA Information

Before creating As-Is or To-Be architecture views, data must exist in analysis repositories. These include:

Business Information

- Goals
- Strategies
- Processes
- Organization units

Application Information

- Software systems
- Interfaces
- Integration points

Data/Information

- Data assets
- Data flows
- Master data definitions

Technology Information

- Infrastructure
- Platforms
- Device inventories

Common Analysis and Documentation Artifacts

Before moving into modeling or views, professionals create artifacts such as:

- ✓ **Architecture Descriptions** - written explanations of how parts of the enterprise work together
- ✓ **Requirements Logs** - what the architecture must satisfy and why
- ✓ **Rationales for Decisions** - documented reasons behind architectural choices
- ✓ **Glossaries and Taxonomies** - shared definitions to avoid misunderstanding
- ✓ **Relationship Matrices** - show how business, applications, data, and technology connect
- ✓ **Gap Analyses** - identify what exists versus what is missing
- ✓ **Stakeholder Maps** - who is affected, involved, or accountable

These artifacts are the raw materials that later become part of As-Is and To-Be views.

Without them, architects cannot reliably produce architecture viewpoints.

Risks of Poor Analysis & Documentation

Poor analysis and documentation lead to serious problems:

- ❌ Incomplete architecture knowledge
- ❌ Ambiguous decisions and lack of traceability
- ❌ Conflicting interpretations across teams
- ❌ Failed projects due to wrong assumptions
- ❌ Lost organizational memory when staff leave

SUMMARY

- Enterprise Architecture relies on structured **analysis** to understand the enterprise as a whole
- **Documentation** captures architectural knowledge for communication, planning, and governance
- Analysis identifies relationships, gaps, dependencies, and impacts across the enterprise
- Good EA documentation must be complete, consistent, traceable, and maintainable
- EA frameworks provide structure for analysis and documentation but do not perform them automatically
- Strong analysis and documentation are essential foundations for developing As-Is and To-Be architecture views

THANK YOU!